CrossAccent-TTS: Cross-Lingual Accent-Intensity Controllable Text-to-Speech via Disentangled Speaker and Accent Representations

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Abstract

Accent conversion and controllability remain fundamental challenges in cross-lingual text-to-speech (TTS), particularly for low-resource and phonetically diverse Indic languages. While recent large language model (LLM)-based TTS systems exhibit strong cross-lingual generalization, they provide limited explicit control over accent characteristics and intensity. In this paper, we propose CrossAccentTTS, a framework that enables both accent control and conversion while preserving speaker identity. Specifically, we introduce an Accent Intensity Controller (AIC) that injects weighted language embeddings into the accent subspace, allowing smooth interpolation between accents and fine-grained modulation of accent strength at inference time. Experiments on the Indic Multilingual and L2-arctic datasets demonstrate that CrossAccent-TTS achieves precise and consistent control of accent intensity, outperforming strong baselines in accent similarity and controllability, while maintaining speaker similarity and naturalness.

Index Terms: LLM based TTS, Controllable Speech Synthesis, Accent, Neural Codecs

1. Introduction

Speech accent refers to systematic variations in phonemes, rhythm, intonation, and linguistic structure, often providing cues about a speaker's background, such as their geographic origin or native language. However, disentangling accent from other speaker-specific attributes, such as pitch range, timbre, and vocal-tract characteristics, remains challenging, as accent is inherently embedded within individual speech patterns. Although accented TTS systems have numerous real-world applications, such as voice dubbing and conversational bots, they have not been a primary focus in mainstream TTS research. The ability to reproduce and control specific accents for a given speaker enables more authentic and personalized speech synthesis. For instance, in dubbing or voice acting, preserving a character's L2 accent, i.e., the foreign accent that arises when speaking a non-native language, enhances realism and audience immersion. Conversely, overly strong or uncontrolled accents may reduce intelligibility, underscoring the need for fine-grained control over accent intensity to balance authenticity, clarity, and expressiveness.

While accent-controllable TTS remains underexplored, recent advances in LLM-based neural codec TTS have demonstrated scalable and flexible speech synthesis frameworks. These approaches combine the language modeling capabilities of LLMs with discrete acoustic representations derived from neural speech codecs. In this paradigm, speech is first encoded into a sequence of discrete acoustic tokens using a neural codec. A large language model is then trained to autoregressively generate these tokens conditioned on text, speaker identity, accent, or other controllable attributes. The generated token sequence is finally decoded by the decoder to reconstruct the final waveform.

To achieve accent-controllable TTS, we propose a Qwen-2.5-based model that uses a neural speech codec (Neucodec) to convert audio into discrete acoustic tokens. The model leverages the generative capability of LLMs to produce accent-conditioned acoustic representations while maintaining speaker identity. In contrast, we address the problem of controllable accent generation in text-to-speech, with particular emphasis on Indic languages, which remain low-resource and underexplored in this context. We additionally validate our method on foreign English accents to demonstrate generalization. Our contributions are summarized as follows:

1. We introduce an Accent Intensity Controller that enables continuous accent modulation at inference time without requiring accent-specific training data.
2. We propose an Accent Suppression Module that promotes adversarial disentanglement of accent from speaker and style representations, thereby preserving speaker similarity.
3. We evaluate our method on the Indic Multilingual and L2 ARCTIC datasets, demonstrating consistent accent controllability across Indian and foreign accents.

The rest of the paper is organized as follows: Section 1 covers related work, followed by Section 3, where we describe the proposed method. Section 4 details the training procedure and experiments, and finally, Section 5 concludes the study.

2. Related Works

Early Text-to-Speech (TTS) systems primarily focused on improving naturalness and speaker similarity, with limited attention given to accent modeling. Traditional approaches [1] attempted to handle accent and speaker variation by conditioning multi-speaker TTS systems on speaker embeddings extracted using a speaker-verification encoder, which generates a fixed-dimensional representation from a short reference speech sample. These embeddings are then used to condition an acoustic model and a neural vocoder, enabling speech synthesis in the voices of both seen and unseen speakers. Since accent information is implicitly embedded within the learned speaker representations, such methods capture accent characteristics indirectly but do not provide explicit control over accent during inference. Other than this, Codec based TTS[2],[3],[4] initially focused on extracting semantic tokens. More recently, single-stream acoustic tokenization approaches have been introduced [5],[6], which directly encode both linguistic and acoustic information, simplifying downstream TTS models and improving

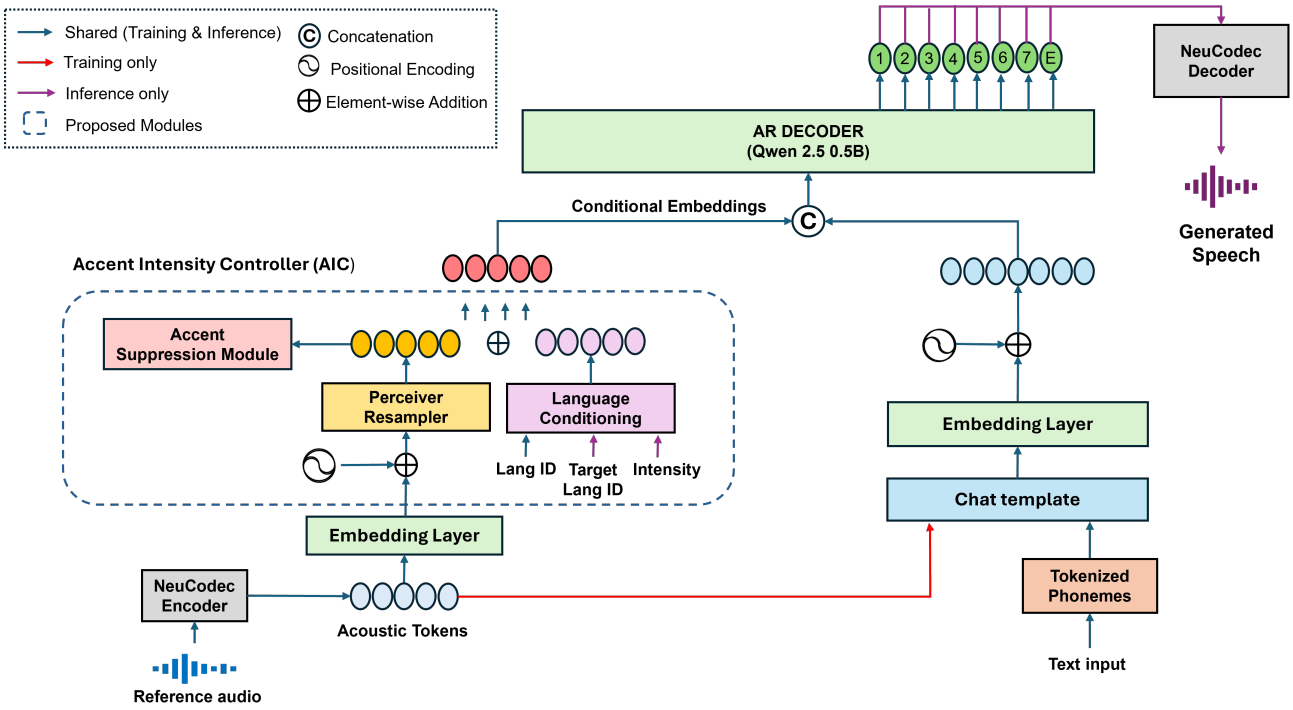

Figure 1: *Model architecture of the proposed CrossAccent TTS*

synthesis quality.

In the context of accent-controllable TTS, [7] explored Indic-language-accented TTS, reducing L2 accent leakage by disentangling accent and speaker representations. However, comparable studies on accent controllability for Indic languages remain limited. To establish strong baselines for Indian languages, we adopt IndicF5[8] and XTTS-v2 [9], which have demonstrated competitive performance on Hindi and other related Indic languages. For English, [10] demonstrated accent control using large-scale datasets and data augmentation strategies. However, such approaches rely heavily on abundant labeled data and are not directly applicable in low-resource Indic settings. To evaluate under controlled low-resource conditions, we use the L2-ARCTIC dataset [11] and compare against CVAE [12] and GST-based [13] baselines. Closest to our inspiration, [14] proposed expanding language embeddings across all acoustic tokens to suppress L2 accent leakage in cross-lingual synthesis. We build on this direction by extending token-level language conditioning to accent-agnostic representations derived from speech codecs.

3. Proposed Methodology

This section describes the proposed Cross-Accent TTS framework. An overview of the architecture is illustrated in 1. The system consists of four main components: (i) speech tokenization using Neucodec, (ii) a Perceiver Resampler for speaker and style encoding with a fixed-length bottleneck, (iii) adversarial suppression of accent and language features, and (iv) explicit language embedding conditioning for controllable accent rendering, followed by auto-regressive acoustic token generation and Neucodec-based waveform synthesis.

3.1. Speech Tokenization with Neucodec

Neucodec [15] is a finite scalar quantization (FSQ)–based neural acoustic speech codec that operates at 50 tokens per second with 16 bits per token. It takes a 16 kHz input and reconstructs speech at 24 kHz using an upsampling decoder. The FSQ encoding scheme also provides improved robustness to bit-level errors. Ground-truth speech is first converted into discrete acoustic tokens using the Neucodec encoder. This encoder maps raw audio waveforms into a sequence of discrete latent codes that preserve perceptual speech quality while providing a compact token-based representation suitable for sequence modeling. These acoustic tokens are used to extract both speaker and style from the reference speech and to train acoustic token generation. During training, a random chunk of acoustic tokens is sampled from each reference utterance. This random chunking strategy reduces overfitting to utterance-level linguistic content and encourages the Perceiver Resampler to focus on speaker-related characteristics rather than specific phonetic sequences.

3.2. Accent Intensity Controller (AIC)

The Accent Intensity Controller learns accent-agnostic speaker features while modeling accent features independently. Both are integrated as conditional embeddings to support explicit control over accent intensity.

3.2.1. Perceiver Resampler

The sampled acoustic token embeddings are fed into a Perceiver Resampler to obtain fixed-length speaker and style embeddings. Perceiver maps variable-length token sequences into a predefined set of latent vectors using cross-attention followed by self-attention layers. Before entering the encoder, learnable positional encodings are added to the reference token embed-

dings to preserve temporal structure. The fixed-length bottleneck constrains the information capacity of the input acoustic embeddings, encouraging the model to retain speaker-specific characteristics such as timbre and vocal quality, forming compact speaker and style embeddings of shape (B, N_s, d). Since these embeddings are used to condition the autoregressive decoder to predict acoustic tokens from text, the bottleneck encourages the model to focus on speaker-related features while discarding linguistic and phonetic variations.[16]

3.2.2. Adversarial learning with Accent Suppression Module

To further suppress residual accent and language information in the speaker and style embeddings, we employ adversarial training with a gradient reversal layer (GRL). An auxiliary classifier is trained to predict the language or accent label from the speaker and style embeddings. During backpropagation, GRL inverts the gradient signal flowing from the classifier to the Perceiver Resampler, encouraging the encoder to remove accent- or language-discriminative information from the speaker and style embeddings. Formally, the Perceiver Resampler is optimized to minimize the TTS reconstruction loss while maximizing the classification error of the auxiliary accent/language classifier. This adversarial objective drives the encoder to produce accent- and language-invariant speaker and style embeddings during inference.

3.2.3. Accent Control using Explicit Language Conditioning

To explicitly control accent and language, we introduce a learned language embedding for each supported language or accent. A simple learnable embedding table is used to obtain a language embedding of shape $(B, 1, d)$. This embedding is then expanded across all latent slots of the speaker and style representations and added to each speaker and style embedding, yielding a combined representation of shape (B, N_s, d). This expansion and addition mechanism increases the relative influence of language conditioning and enables the model to explicitly control linguistic and accent-related factors independently of the speaker and style.

3.3. Autoregressive Acoustic Token Generation

The combined speaker–language representation is provided as conditional input to an autoregressive decoder based on the Qwen architecture. We use Qwen 2.5 (0.5B) [17] with the same configuration as the original model. The decoder jointly conditions on text token embeddings and the combined speaker–language embeddings to predict sequences of acoustic tokens. This autoregressive formulation allows the model to capture long-range dependencies and produce coherent acoustic token sequences conditioned on both linguistic content and controllable accent information. The final training objective is defined as:

$$L_{\text{total}} = L_{\text{decoder}} + \lambda_{\text{GRL}}\, L_{\text{GRL}} \tag{1}$$

where L_{decoder} is the autoregressive acoustic token prediction loss, and L_{GRL} is the adversarial classification loss used for accent and language suppression and λ_{GRL} is a weighting factor that controls the relative contribution of the GRL loss in the total training objective.

4. Training and Evaluation Setup

This section describes the experimental setup, evaluation metrics, and results to assess the proposed model's performance.

4.1. Datasets

We conduct experiments on two datasets, Indic Multilingual and L2 ARCTIC, to evaluate performance in low-resource and second-language (L2) accent scenarios.

4.1.1. Indic Multilingual Dataset

We use a multilingual speech corpus comprising Hindi, Telugu, Tamil, Bengali, Marathi, and English. The dataset contains 636 hours of internal in-house Indic speech data and 350 hours of custom-split speech from the Emilia Yodas dataset [18], totaling approximately 986 hours of speech. All audios are resampled to 16 kHz for Neucodec encoding. Language labels are used to explicitly condition language embeddings and to perform adversarial training.

4.1.2. L2 Arctic Dataset

The L2 Arctic corpus comprises 27 hours of non-native English speech from 24 speakers, with six distinct accent backgrounds. The dataset provides aligned text and speech with annotated speaker and native-language labels, making it suitable for evaluating accent neutrality and for controllable accent rendering in L2 English synthesis.

4.2. Training Setup

The proposed model is trained end-to-end using paired text and speech data. The text is converted to International Phonetic Alphabets (IPA) and tokenized using a shared tokenizer that maps speech tokens. The Perceiver Resampler uses $N_s = 32$ latent slots and embedding dimension $d = 768$. The autoregressive decoder is initialized from Qwen 2.5 (0.5B) and fine-tuned for acoustic token generation. The auxiliary accent/language classifier is trained jointly with the main model using the accent suppression module. The total training loss is defined as shown in Eq. 1, where λ_{GRL} controls the strength of adversarial accent and language suppression.

We set $\lambda_{\text{GRL}} = 0.1$ in all experiments. The decoder is first trained on the full 986-hour multilingual dataset for 5 epochs, then fine-tuned on the L2 Arctic dataset with a reduced learning rate for 4 additional epochs. During inference, the reference speech is used to extract speaker and style information, and multiple language embeddings can be linearly combined to control accent intensity. Specifically, given language embeddings e_{lang_1} and e_{lang_2}, a weighted combination is used,

$$\lambda e_{\text{lang}_1} + (1 - \lambda) e_{\text{lang}_2} \tag{2}$$

where $\lambda \in [0, 1]$ controls the relative contribution. The predicted acoustic tokens are passed through the Neucodec decoder to reconstruct the time-domain speech waveform.

4.3. Evaluation Metrics

We evaluate the model's performance using both objective and subjective metrics.

4.3.1. Objective Metrics

To obtain accent embeddings for evaluation, we use the GenAID model originally introduced in the Accent Box [11] for L2 English accent analysis. For Indic languages, we fine-tune the same model using approximately 100k samples from 11 Indian languages and around 3,300 speakers from the Indic Voices dataset [19] for two epochs. Accent embeddings extracted from

this fine-tuned model are used for Indic accent evaluations. We report the following objective measures:

- **Accent Similarity:** Measures how closely synthesized speech matches the target accent. We compute cosine similarity between the accent embeddings of generated speech and those of ground-truth speech in the target accent. Higher values indicate better accent rendering.
- **Accent Leakage:** Measures how much accent information from the reference audio is preserved when synthesizing in a different target accent. We compute the cosine similarity between the accent embeddings of the reference and generated audio. Lower values indicate better suppression of source accent.
- **UTMOS [20]:** This metric is used to assess overall speech quality.
- **Speaker Similarity:** Computed using a pretrained speaker verification model (Resemblyzer) [21] to assess speaker identity preservation.

4.3.2. Subjective Metrics

Subjective listening tests are conducted using the Mean Opinion Score (MOS) to evaluate how closely the synthesized speech matches the target accent. Listening tests are conducted with 20 participants (aged between 23 to 35 with no known hearing impairments).

4.4. Evaluation Results

4.4.1. Objective Results

Table 1 and Table 2 report the objective evaluation results on the Indic multilingual and L2 Arctic datasets. The proposed method achieves reduced accent leakage compared to baselines, demonstrating more effective suppression of source accent in accent-neutral synthesis. When explicit language embeddings are applied, accent similarity increases in a controlled manner, confirming the effectiveness of the proposed accent control mechanism. The proposed method also maintains speaker similarity scores comparable to the external speaker embedding baseline, indicating that speaker identity is preserved despite adversarial accent suppression. The UTMOS results further show that overall speech quality is maintained or improved relative to baseline systems.

Table 1: *Objective evaluation results on Indian accent conversion (↑ higher is better, ↓ lower is better).*

Model	UTMOS ↑	AccLeak ↓	AccSim ↑	SpkSim ↑
IndicF5	2.817	0.312	0.312	**0.843**
XTTS_v2	3.168	0.284	0.284	0.832
Proposed	**3.181**	**0.203**	**0.371**	0.842

Table 2: *Objective evaluation results on L2 Arctic dataset with English accents (↑ higher is better, ↓ lower is better).*

Model	UTMOS ↑	AccLeak ↓	AccSim ↑	SpkSim ↑
CVAE-L	2.810	0.487	0.612	0.677
CVAE-NL	2.714	0.530	0.491	0.673
GST	3.044	0.544	0.670	**0.732**
Proposed	**4.001**	**0.439**	**0.686**	0.693

4.4.2. Subjective Results

Figure 2 and Figure 3 represent the subjective listening test results for Indian accents and foreign accents, respectively. The proposed method achieves higher MOS scores for accent similarity compared to baseline systems. In accent-controlled settings, listeners rate the synthesized speech as more similar to the target accent, confirming that weighted language embedding enables perceptually meaningful accent control.

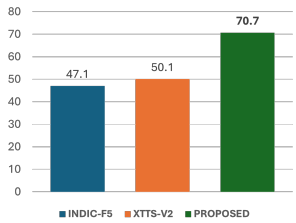

Figure 2: *Accent Similarity MOS of Indic Accents*

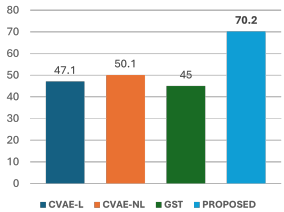

Figure 3: *Accent Similarity MOS of Foreign Accents*

4.5. Accent-intensity Control Analysis

To assess accent-intensity control, we synthesize samples using cross-lingual reference audio, scaling the accent intensity to 0, 0.3, 0.6, and 1.0. Accent embeddings from the synthesized speech are extracted using GenAID, and accent similarity scores are computed with the mentioned intensity scores. From the Figure 4, it is evident that as accent intensity increases, the accent similarity score also rises, demonstrating effective and controllable accent modulation.

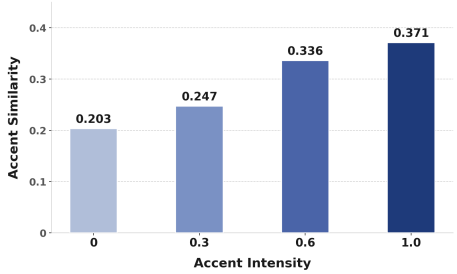

Figure 4: *Effect of Accent Intensity on Accent Similarity score*

5. Conclusion

This paper presents CrossAccent TTS, a cross-lingual, accent-controllable text-to-speech framework that enables both accent conversion and continuous modulation of accent intensity while preserving speaker identity. The proposed Accent Intensity Controller injects weighted language embeddings into disentangled representations, allowing smooth interpolation across accents and dynamic control during inference. Extensive objective and subjective evaluations demonstrate that the proposed approach consistently outperforms strong baselines in accent similarity and controllability, while maintaining speaker similarity and naturalness. The effectiveness of the learned accent representations is further supported by accent-intensity control analysis. Overall, the proposed framework provides a practical and scalable solution for accent-controllable speech synthesis in low-resource and multilingual settings.

6. Generative AI Use Disclosure

Generative AI tools were used solely for language refinement, grammar correction, and improving the overall clarity and readability of the manuscript. These tools assisted in polishing and structuring the text but were not used to generate or design the core research ideas, methodology, experimental setup, analysis, or conclusions presented in this work. All scientific contributions, technical implementations, and interpretations were developed and validated by the authors. In accordance with policy guidelines, no generative AI system is listed as a co-author, and all authors take full responsibility and accountability for the content and integrity of this paper.

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